

UndoProof

Prove the way back

PostgreSQL rollback rehearsal by Shivam Gupta

The rollback succeeds. The new order disappears.

A release can pass forward tests while its recovery path breaks the previous application contract or erases data written after deployment. UndoProof makes that failure reproducible before a migration ships.

HOW IT WORKS

A contract supplies synthetic fixtures, migration and rollback SQL, application queries and a data invariant. UndoProof executes real PostgreSQL through PGlite and exports the SQL, errors and before-and-after rows. IBM Bob added and ran 41 regression tests for source-derived checkout contracts, data loss and unchanged repair invariants.

THE IMPORTANT CHECK

Write a new order *after* the migration. Roll back. Compare the customer-visible data. A safe recovery can retain a compatible expanded schema when the application rolls back, instead of restoring an obsolete snapshot.

FOR DEVELOPERS

Review the failed release state, inspect a repair, and rerun the same contract. The browser and command-line workflow share an execution engine. Rehearsals need no production credentials or paid model inference.

COMMERCIAL HYPOTHESIS

Free MIT local runner. A proposed \$99/month team workspace would add shared policies and retained evidence. Initial audience: platform leads at SaaS teams. Pricing and demand require customer validation.

SCOPE

Results cover the supplied SQL and fixtures in embedded PostgreSQL. They do not establish full application behavior, concurrent transaction behavior, production lock safety, or production parity.

undoproof.web.app

Source: github.com/shi1720/ibm-bob

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